A Lightweight Trust Management Framework in the Networking Layer of IoT Devices.

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Project Proposal Report

Kaluarachchi D.N.

B.Sc. (Hons) in Information Technology Specializing in Computer Systems & Network Engineering

Department of Computer Systems Engineering

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Declaration

I declare that this is my own work, and this proposal does not incorporate without acknowledgment any material previously submitted for a degree or diploma in any other university or Institute of higher learning and to the best of our knowledge and belief it does not contain any material previously published or written by another person except where the acknowledgment is made in the text.

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The above candidates are carrying out research for the undergraduate dissertation under my supervision.

Signature of the Supervisor: ..... Date 24/8/25

Signature of the Co-Supervisor: ..... Date 24/08/2025

Acknowledgement

Working on this final-year research project has been an immensely enriching academic experience for me. Through the process of developing my research proposal, I have gained valuable insights into the research process, including how to formulate research questions, design suitable methodologies, and anticipate potential challenges.

I would like to express my heartfelt gratitude to my supervisor, Ms. Shashika Lokuliyana for her invaluable guidance, feedback, and support throughout this project. Her constant input and insightful suggestions are greatly appreciated. I would also like to express my gratitude to my co-supervisor, Mr. Anuradha Jayakody, for his unwavering guidance, feedback, and support throughout the project. I owe a debt of gratitude to all the lecturers at SLIIT as well.

I look forward to continuing to develop my research skills and contribute meaningful insights through this project.

Abstract

The rapid proliferation of Internet of Things (IoT) devices utilizing the Routing Protocol for Low-Power and Lossy Networks (RPL) has introduced significant security challenges. Traditional cryptographic methods, while effective in conventional networks, are often impractical in IoT due to energy limitations and high computational overhead [2], [4]. Moreover, insider threats and protocol-level vulnerabilities such as selective forwarding and sinkhole attacks further reduce the reliability of these systems [3], [6]. These shortcomings highlight the urgent need for lightweight, adaptive, and energy-efficient trust management frameworks.

To address these challenges, this research proposes a Lightweight Trust Management Framework specifically designed for RPL-based IoT networks. The framework integrates four key components: an Energy-Efficient Trust Computation Engine, a Trust-Aware RPL Routing Mechanism, a Secure Attack Detection Module, and Isolation, Mitigation, and Recovery mechanisms. The novelty lies in the adaptive scheduling of trust computations based on battery status [16], multi-metric trust evaluation including residual energy and temperature [1], [7], and machine learning-driven behavioral analysis for attack detection [8], [12]. These design choices ensure low overhead while maintaining high accuracy and resilience against evolving threats.

The expected impact of this research is threefold: (1) extending IoT network lifetime by reducing energy overhead through adaptive trust scheduling, (2) enhancing routing reliability by embedding trust metrics directly into RPL objective functions, and (3) improving resilience through real-time detection and rapid recovery mechanisms. Simulation results in Cooja and NS-3 are anticipated to demonstrate improvements of up to 40% in energy efficiency, 25% in packet delivery reliability, and sub-120ms response times compared to existing solutions. Ultimately, this work aims to establish a paradigm shift from isolated, reactive security solutions to a proactive, integrated, and sustainable trust management framework suitable for deployment in critical IoT applications such as disaster management, smart cities, and industrial automation.

Keywords — IoT Trust Management, RPL Security, Energy-Aware Scheduling, Lightweight Framework, Adaptive Routing

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1.0 INTRODUCTION

The Internet of Things (IoT) has revolutionized modern computing by connecting billions of devices worldwide, creating unprecedented opportunities for smart applications across various domains including healthcare, industrial automation, smart cities, and environmental monitoring. However, the proliferation of IoT devices has introduced significant security challenges, particularly in networks utilizing the Routing Protocol for Low-Power and Lossy Networks (RPL), which serves as the de facto standard for routing in IoT environments.

RPL-based IoT networks face unique security vulnerabilities due to their resource-constrained nature, distributed architecture, and wireless communication medium. Traditional security mechanisms, primarily based on cryptographic approaches, often prove inadequate for IoT environments due to their computational overhead, energy consumption, and inability to address insider attacks [11]. Consequently, trust management has emerged as a promising alternative security paradigm that can effectively mitigate internal threats while maintaining the lightweight characteristics essential for IoT deployments [2].

Current trust management systems for IoT networks suffer from several critical limitations including static trust computation schedules, inadequate integration with routing protocols, high false positive rates in attack detection, and inefficient response mechanisms [6]. These limitations become particularly pronounced in RPL-based networks where routing decisions significantly impact network performance and security [3].

The energy constraint challenge is especially critical in IoT deployments where nodes operate on limited battery power for extended periods. Recent studies have demonstrated that traditional trust computation systems can consume up to 15-20% of total node energy, significantly reducing network lifetime [17]. This energy overhead becomes even more problematic when nodes are deployed in remote or inaccessible locations where battery replacement is impractical [4].

Recent advances in trust-aware RPL routing have shown promising results. The development of novel reliable and trust objective functions (RSOF) has demonstrated significant improvements in parent selection processes by incorporating multiple trust metrics including residual energy, node failure rates, and extended RPL node trustworthiness [1]. Similarly, fuzzy logic-based whale optimization algorithms for trust-aware routing (FSRW) have achieved enhanced security in IoT-based healthcare applications through intelligent trust computation and routing optimization [4]. Trust-aware RPL routing protocols (TARRP) have further contributed by providing reliable routing through optimized trust-based topology creation and malicious node identification [3].

The integration of lightweight trust mechanisms into RPL routing has addressed computational efficiency challenges inherent in resource-constrained environments. Trust-based Hybrid

Cooperative RPL protocols (THC-RPL) have achieved over 99.85% detection accuracy against Sybil attacks while maintaining energy efficiency and reducing communication overhead by 10% compared to existing approaches [2,5]. These protocols demonstrate the feasibility of implementing sophisticated trust mechanisms without compromising network performance in IoT environments.

Mobile IoT environments have introduced additional challenges requiring mobility-aware trust mechanisms. Trust and mobility-based protocols (SMTrust) have addressed RPL rank and blackhole attacks in dynamic environments, demonstrating superior performance compared to standard objective functions through consideration of node mobility metrics and behavioral analysis [6,7]. These advances establish the foundation for comprehensive trust-aware routing that adapts to both static and mobile IoT deployment scenarios.

Real-time attack detection capabilities have evolved significantly with machine learning integration. Advanced neural network architectures incorporating threat intelligence have achieved classification accuracy rates exceeding 99.89% for detecting cyber attacks in IoT-WSN devices, specifically addressing man-in-the-middle threats through hidden and connected layer architectures [8]. Computational intelligence techniques utilizing particle swarm optimization and feature selection methods have enhanced intrusion detection system performance while reducing computational complexity for wireless sensor networks [9]. Hybrid machine learning models addressing class imbalance and dimensionality reduction challenges have demonstrated superior performance in wireless sensor network intrusion detection, achieving optimal balance between accuracy and real-time processing requirements [10].

Machine learning and deep learning techniques for IoT network anomaly detection have achieved significant breakthroughs in identifying various attack types including denial of service, data manipulation, and unauthorized access attempts [11]. Smart IoT security frameworks utilizing lightweight machine learning techniques have successfully implemented multi-class attack detection in resource-constrained IoT networks while maintaining minimal computational overhead [14]. Advanced hybrid deep learning models combining AutoEncoder feature extraction with XGBoost classification (AEXB) have demonstrated 97.24% accuracy in real-time detection and prevention of man-in-the-middle cyber attacks [12].

The evolution toward comprehensive intrusion detection frameworks has emphasized multi-access edge computing integration and diverse dataset utilization for enhanced generalizability [13]. Intrusion detection systems specifically designed for IoT environments have successfully leveraged machine learning approaches to address multiple attack vectors while considering the unique constraints of wireless sensor network deployments.

Self-healing and recovery mechanisms represent the newest frontier in IoT security research. Runtime self-healing security systems (Antilizer) have introduced distributed trust model construction through

network overhearing, enabling autonomous adaptation of communication decisions and collaborative routing to avoid compromised network regions [15]. Intelligent fault detection systems incorporating machine learning with Flying Fox Optimization Algorithm (FFOA) have achieved average recovery times of 120 milliseconds with 98.5% connectivity restoration ratios, demonstrating the practical feasibility of real-time self-healing mechanisms [16].

Autonomous self-aware and adaptive fault tolerant routing techniques (ASAART) have demonstrated superior performance in wireless sensor networks through combined continuous and slotted prioritized transmission mechanisms, achieving enhanced energy efficiency and routing performance compared to traditional approaches [17]. Hardware-based self-healing approaches utilizing reconfigurable technologies have provided dynamic configuration adjustment capabilities for automatic failure recovery at the node level [18].

This research builds upon the extensive foundation of recent trust management advances while introducing a novel four-component integrated framework specifically designed for energy-constrained RPL-based environments. Unlike existing solutions that address individual security aspects in isolation, our proposed framework introduces several key innovations through systematic integration:

Energy-Efficient Trust Computation Engine: The framework dynamically adjusts trust computation frequency based on real-time battery status and network conditions, representing the first comprehensive integration of energy awareness directly into trust computation scheduling. This approach builds upon recent energy optimization work demonstrated in RSOF and FSRW while extending it to multi-metric behavioral analysis contexts [1,4].

Trust-Aware RPL Routing Mechanism: While existing solutions typically focus on standard objective functions, our approach integrates trustworthiness and security factors directly into RPL routing decisions through dynamic parent selection algorithms and secure route path broadcasting, advancing beyond the single-metric approaches demonstrated in THC-RPL and SMTrust literature [2,6].

Secure Attack Detection Mechanism: The system incorporates real-time behavioral analysis with statistical models for attack detection, implementing adaptive detection thresholds that minimize false positives in dynamic environments, building upon recent machine learning advances demonstrated in IoT intrusion detection research [8,9,10,11].

Isolation, Mitigation and Recovery Mechanisms: The framework includes specialized response mechanisms for threat isolation, real-time attack mitigation, and trust recovery for compromised nodes, extending current self-healing research to comprehensive attack response scenarios demonstrated in Antilizer and intelligent fault detection systems [15,16].

The novelty of this research lies in the unified integration of energy-aware trust computation, behavioral pattern analysis, adaptive routing mechanisms, and intelligent response systems in a comprehensive four-component framework. This represents a paradigm shift from reactive, isolated security mechanisms to proactive, energy-conscious, and fully integrated trust management that can sustain long-term IoT network operations while maintaining robust security coverage across all network's operational aspects.

1.1 Background and Literature Review

1.1.1 Background

1. Internet of Things (IoT) Evolution and Applications

The Internet of Things (IoT) represents a paradigmatic shift in computing, connecting billions of heterogeneous devices worldwide to create intelligent, interconnected systems. IoT networks enable unprecedented opportunities for smart applications across diverse domains including healthcare monitoring, industrial automation, smart city infrastructure, environmental sensing, and disaster management systems. The rapid proliferation of IoT devices, projected to exceed 75 billion connected devices by 2025, has fundamentally transformed how we interact with and monitor our physical environment.

IoT deployments are characterized by several unique characteristics that distinguish them from traditional networks: resource constraints (limited processing power, memory, and energy), heterogeneous device types, wireless communication dependencies, and distributed autonomous operation. These characteristics create both opportunities for innovative applications and significant challenges for security, reliability, and sustainability.

2. RPL Protocol Fundamentals

The Routing Protocol for Low-Power and Lossy Networks (RPL) serves as the de facto standard for routing in IoT environments, specifically designed to address the unique requirements of resource-constrained networks. RPL operates by constructing Directed Acyclic Graphs (DAGs) rooted at border routers, enabling efficient multi-hop communication between IoT devices and external networks.

2.1 RPL Architecture and Operations

RPL utilizes objective functions (OFs), such as OF0 and MRHOF, to determine optimal parent selection and routing paths based on metrics including hop count, Expected Transmission Count

(ETX), residual energy, and link quality indicators. The protocol employs three primary control messages: DIO (DODAG Information Object) for downward route establishment, DAO (DODAG Advertisement Object) for upward route maintenance, and DIS (DODAG Information Solicitation) for topology discovery.

Recent advances in RPL optimization have focused on enhancing objective functions to incorporate trust and security considerations. For example, Zhou et al. proposed the Reliable and Secure Objective Function (RSOF), which integrates multiple trust metrics including residual energy, node failure rates, and extended trustworthiness, thereby improving parent selection and reliability in RPL-based networks [1]. Similarly, Jiang and Liu analyzed trust-based defenses against selective forwarding attacks, highlighting the effectiveness of integrating trust into routing decisions [3].

Other works have introduced lightweight trust-enabled RPL variants such as THC-RPL, which effectively detects Sybil attacks with high accuracy while minimizing energy consumption [2]. Additionally, Singh and Kaur proposed FSRW, a fuzzy logic-based whale optimization algorithm that enhances trust-aware routing in IoT-based healthcare applications [4]. Verma and Sahu also presented a trust-based secure routing model against RPL attacks, further validating the role of trust metrics in strengthening RPL [7].

These trust- and security-aware objective functions represent significant improvements over traditional RPL implementations, demonstrating reduced false positives, improved packet delivery ratios, and more reliable routing in resource-constrained IoT environments [1–4,7].

2.2 RPL Security Challenges

Standard RPL implementations are susceptible to various attack vectors that exploit the protocol's inherent trust assumptions. Common attacks include rank attacks (manipulating node positions in the DODAG), sinkhole attacks (attracting traffic through false advertisements), selective forwarding attacks (dropping specific packets), version number attacks (forcing topology reconstruction), and insider attacks (compromised legitimate nodes).

Research has shown that these vulnerabilities can significantly impact network performance and security. Secure IoT routing approaches addressing selective forwarding attacks and trust-based defenses have demonstrated that these attacks can be particularly devastating in resource-constrained environments, necessitating lightweight defense solutions [3]. The challenge lies in developing security mechanisms that can effectively detect and mitigate these attacks while maintaining the energy efficiency essential for IoT deployments.

3. Trust Management Fundamentals

Trust management has emerged as a complementary security paradigm that addresses limitations of traditional cryptographic approaches in IoT environments. Unlike cryptographic methods that focus

on preventing unauthorized access, trust management evaluates the reliability and trustworthiness of network participants based on their observed behavior patterns.

3.1 Trust Computation Models

Trust computation in IoT networks typically involves multiple dimensions including direct trust (based on personal interactions), indirect trust (based on recommendations), and context-specific trust (considering operational conditions). Trust-based hybrid cooperative RPL protocols have demonstrated the effectiveness of multi-dimensional trust evaluation incorporating peer-to-peer communication quality, service reliability, and contextual information [2,5].

Recent advances have introduced sophisticated trust computation mechanisms that achieve remarkable detection accuracy. These systems have demonstrated over 99.85% detection accuracy against sophisticated attacks such as Sybil attacks, while maintaining energy efficiency through optimized computation scheduling and lightweight trust-enabled routing mechanisms.

3.2 Behavioral Pattern Analysis

Modern trust systems have evolved beyond simple reputation mechanisms to incorporate comprehensive behavioral pattern analysis. This approach considers multiple behavioral indicators including communication patterns, energy consumption characteristics, response times, and collaboration effectiveness. Trust and mobility-based protocols (SMTrust) have shown that incorporating mobility metrics and behavioral analysis significantly improves security against RPL attacks in dynamic environments [6,7].

Advanced behavioral pattern recognition has benefited significantly from machine learning integration. Neural network architectures incorporating threat intelligence have achieved classification accuracy rates exceeding 99.89% for detecting behavioral changes in IoT devices, particularly in identifying nodes that exhibit suspicious behavior patterns [8].

4. Energy Constraints in IoT Networks

Energy limitation represents one of the most critical challenges in IoT deployments, particularly in scenarios where devices operate on battery power for extended periods without maintenance access. This constraint significantly impacts the design and implementation of security mechanisms, including trust management systems.

4.1 Energy Consumption Patterns

Research has demonstrated that security mechanisms can consume substantial portions of total node energy. Traditional trust computation systems consume 15-20% of total node energy through static computation schedules, leading to premature network degradation and operational failure [17]. This

energy consumption becomes particularly problematic in remote deployment scenarios where battery replacement is impractical or impossible.

Recent advances in energy-aware systems have focused on developing intelligent scheduling mechanisms. Intelligent fault detection and self-healing mechanisms utilizing machine learning with Flying Fox Optimization Algorithm (FFOA) have achieved remarkable energy efficiency while maintaining 98.5% connectivity restoration ratios with average recovery times of 120 milliseconds [16].

4.2 Power Optimization Strategies

Contemporary research has introduced various approaches to address energy constraints in security systems. Self-healing security systems (Antilizer) have demonstrated effective runtime self-healing capabilities through distributed trust model construction, enabling autonomous adaptation while minimizing energy consumption [15]. These systems achieve security objectives through network overhearing and collaborative routing mechanisms that avoid compromised network regions.

Autonomous self-aware and adaptive fault tolerant routing techniques (ASAART) have shown superior performance through combined continuous and slotted prioritized transmission mechanisms, achieving enhanced energy efficiency compared to traditional routing approaches [17]. Hardware-based approaches utilizing reconfigurable technologies have provided additional opportunities for dynamic configuration adjustment and automatic failure recovery at the node level [18].

5. Security Challenges in IoT Networks

IoT networks face unique security challenges due to their distributed nature, resource constraints, and wireless communication dependencies. These challenges necessitate specialized security mechanisms that can operate effectively within the constraints of IoT environments.

5.1 Attack Landscape

IoT networks are subject to various attack types that exploit different vulnerabilities. Machine learning and deep learning techniques for IoT network anomaly detection have identified multiple attack categories including denial of service attacks, data manipulation attacks, unauthorized access attempts, and sophisticated insider attacks [11]. Smart IoT security frameworks have demonstrated the need for lightweight machine learning techniques capable of multi-class attack detection while maintaining minimal computational overhead [14].

Advanced attack detection mechanisms have achieved significant success in identifying sophisticated threats. Hybrid deep learning models combining AutoEncoder feature extraction with XGBoost classification (AEXB) have demonstrated 97.24% accuracy in real-time detection and prevention of

man-in-the-middle cyber-attacks [12]. These systems represent significant advances in real-time threat detection capabilities suitable for resource-constrained environments.

5.2 Intrusion Detection Systems

The development of specialized intrusion detection systems for IoT environments has focused on addressing the unique requirements of wireless sensor networks. Computational intelligence techniques utilizing particle swarm optimization and feature selection methods have enhanced intrusion detection system performance while reducing computational complexity [9]. These approaches demonstrate that intelligent feature selection can significantly improve detection efficiency without compromising accuracy.

Intrusion detection systems leveraging multi-access edge computing have provided enhanced capabilities for comprehensive threat detection [13]. Hybrid machine learning models addressing class imbalance and dimensionality reduction challenges have demonstrated superior performance in wireless sensor network intrusion detection, achieving optimal balance between accuracy and real-time processing requirements [10].

6. Current Limitations and Research Motivation

Despite significant advances in individual security mechanisms, current approaches face several critical limitations that motivate the need for integrated, energy-aware trust management frameworks.

6.1 Integration Challenges

Existing trust management systems typically address individual security aspects in isolation, lacking comprehensive integration that can provide holistic security coverage. While systems like fuzzy secure hierarchical routing (FSRW) have demonstrated effectiveness in specific applications [4], there remains a need for unified frameworks that can integrate multiple security mechanisms while maintaining energy efficiency.

6.2 Energy Efficiency Gaps

Current trust computation systems lack sophisticated energy-aware scheduling mechanisms that can adapt to real-time energy availability and network conditions. Despite advances in individual optimization techniques, comprehensive frameworks that balance security effectiveness with energy conservation remain underdeveloped.

7. Research Foundation and Motivation

This research builds upon the extensive foundation of recent advances in trust-aware routing, machine learning-based security, and energy optimization to develop a comprehensive lightweight trust management framework. The integration of energy-efficient trust computation, behavioral pattern

analysis, adaptive routing mechanisms, and intelligent response systems represents a significant advancement over existing isolated security approaches.

The motivation for this research stems from the critical need for sustainable security solutions that can operate effectively in resource-constrained IoT environments while providing comprehensive protection against evolving threats. The proposed framework addresses the identified limitations through systematic integration of proven techniques while introducing novel energy-aware computation scheduling mechanisms that enable long-term sustainable operation.

1.1.2 Literature Survey

1. Introduction

Trust management in Internet of Things (IoT) networks has evolved from simple reputation-based systems to sophisticated multi-metric frameworks that address the unique challenges of resource-constrained environments. This literature review examines the current state of energy-efficient trust computation systems, behavioral pattern analysis techniques, adaptive trust update mechanisms, and energy-aware computation scheduling specifically in the context of RPL-based IoT networks. The review synthesizes findings from recent research to identify gaps and opportunities for advancing energy-efficient trust computation engines.

2. Energy-Efficient Trust Management Systems

2.1 Evolution of Energy-Aware Trust Frameworks

The development of energy-efficient trust management systems has gained significant attention as IoT deployments scale and energy constraints become more critical. Recent research has demonstrated substantial progress in integrating energy considerations directly into trust computation architectures.

Novel approaches to reliable and secure objective functions have emerged, with researchers developing comprehensive frameworks that incorporate multiple trust metrics including residual energy, node failure rates, and extended RPL node trustworthiness [1]. These systems have shown significant improvements in network performance, particularly in terms of average latency reduction and packet delivery ratio enhancement, while maintaining energy efficiency through adaptive parent selection mechanisms.

Trust-based hybrid cooperative RPL protocols have addressed the dual challenge of maintaining security while preserving energy resources [2,5]. These systems evaluate node trustworthiness through lightweight trust-enabled routing mechanisms, achieving over 99.85% detection accuracy against Sybil attacks while maintaining energy efficiency and reducing communication overhead by

approximately 10% compared to existing approaches. The research demonstrates that continuous trust monitoring, while incurring slight energy overhead, results in significant improvements in packet delivery ratio and attack detection capabilities.

2.2 Trust-Aware Routing Integration

The integration of trust mechanisms directly into routing protocols has shown promising results in energy optimization. Secure IoT routing approaches utilizing selective forwarding attacks and trust-based defenses have introduced lightweight defense solutions that achieve high detection accuracy with very limited extra energy usage (approximately 3.4%) [3]. These approaches demonstrate the feasibility of implementing sophisticated security mechanisms without significant energy penalties.

Fuzzy logic integration in trust-based routing has addressed uncertainty challenges inherent in IoT environments [4]. The FSRW approach utilizing fuzzy logic-based whale optimization algorithms for trust-aware routing in IoT-based healthcare has shown effectiveness against various attack types while maintaining energy efficiency through intelligent optimization techniques. Multi-dimensional trust evaluation incorporating contextual information and quality of service metrics has demonstrated superior performance in dynamic IoT environments.

3. Behavioral Pattern Analysis for Trust Evaluation

3.1 Multi-Metric Behavioral Assessment

Contemporary research has moved beyond single-metric trust evaluation toward comprehensive behavioral pattern analysis. Trust-based hybrid cooperative protocols have demonstrated the effectiveness of subjective logic trust models that consider multiple dimensions including peer-to-peer communication, quality of service, and contextual information [2,5]. These systems achieve detection accuracy rates exceeding 99.85% against sophisticated attacks while maintaining energy conservation through optimized computation scheduling.

The consideration of mobility in trust evaluation has introduced additional complexity and opportunities. Trust and mobility-based protocols (SMTrust) have shown that incorporating mobility metrics significantly improves security against RPL rank and blackhole attacks [6,7]. The research demonstrates that mobile IoT environments require specialized trust mechanisms that can adapt to changing node positions while maintaining energy efficiency through context-aware computation.

3.2 Advanced Detection Mechanisms

Recent advances in attack detection have incorporated sophisticated behavioral analysis techniques. Trust-based models for secure routing against RPL attacks have shown effectiveness through comprehensive evaluation of node behavior patterns [7]. These systems provide secure

communication among resource-constrained nodes while enhancing reliable and trustworthy communication through adaptive trust computation.

Machine learning integration has significantly advanced behavioral pattern recognition capabilities. Advanced neural network architectures incorporating threat intelligence have achieved classification accuracy rates exceeding 99.89% for detecting cyber attacks in IoT-WSN devices [8]. The integration of hidden and connected layer architectures enables real-time identification of nodes exhibiting behavioral changes between normal and malicious states, with particular effectiveness in detecting man-in-the-middle attacks.

4. Adaptive Trust Update Algorithms

4.1 Dynamic Computation Strategies

The development of adaptive trust update mechanisms has focused on optimizing computation frequency and resource allocation. Computational intelligence techniques utilizing particle swarm optimization and feature selection methods have enhanced intrusion detection system performance while reducing computational complexity [9]. These approaches demonstrate that intelligent feature selection can significantly improve the efficiency of trust computation without compromising accuracy.

Research in hybrid machine learning models has addressed critical challenges including class imbalance and dimensionality reduction in trust computation systems [10]. The integration of data balancing techniques with dimensionality reduction and advanced classification algorithms has demonstrated superior performance in wireless sensor network trust evaluation, achieving optimal balance between accuracy and real-time processing requirements.

4.2 Real-Time Adaptation Mechanisms

Machine learning and deep learning techniques for IoT network anomaly detection have incorporated sophisticated algorithms to achieve rapid trust updates [11]. These systems demonstrate effectiveness across multiple attack types including denial of service, data manipulation, and unauthorized access attempts, indicating the potential for real-time trust adaptation in resource-constrained environments.

Advanced hybrid deep learning models combining AutoEncoder feature extraction with XGBoost classification (AEXB) have shown 97.24% accuracy in real-time detection and prevention of man-in-the-middle cyber attacks [12]. The research demonstrates that sophisticated trust computation can be achieved while maintaining energy efficiency through optimized algorithm selection and feature extraction techniques, with minimal computational overhead suitable for IoT environments.

5. Energy-Aware Computation Scheduling

5.1 Battery-Conscious Trust Systems

The development of energy-aware computation scheduling has emerged as a critical factor in sustainable IoT deployments. Research has shown that traditional trust computation systems can consume 15-20% of total node energy, necessitating adaptive scheduling mechanisms that balance security requirements with energy conservation [17].

Intelligent fault detection and self-healing mechanisms utilizing machine learning with Flying Fox Optimization Algorithm (FFOA) have achieved average recovery times of 120 milliseconds with 98.5% connectivity restoration ratios [16]. These systems demonstrate the potential for real-time adaptation while maintaining energy efficiency through optimized computation scheduling.

5.2 Power Optimization Strategies

Recent advances in power optimization for trust operations have focused on algorithm efficiency and resource sharing. Self-healing security systems (Antilizer) have introduced distributed trust model construction through network overhearing, enabling autonomous adaptation while minimizing energy consumption [15]. The research demonstrates effective runtime self-healing capabilities with reduced computational overhead.

Autonomous self-aware and adaptive fault tolerant routing techniques (ASAART) have shown superior performance through combined continuous and slotted prioritized transmission mechanisms [17]. These approaches achieve enhanced energy efficiency and routing performance compared to traditional methods, demonstrating the potential for comprehensive energy optimization in trust-aware systems.

6. Conclusion

The literature review reveals significant progress in energy-efficient trust management for IoT networks, with notable advances in behavioral pattern analysis through THC-RPL and SMTrust approaches [2,5,6,7], adaptive algorithms utilizing machine learning techniques [8,9,10,11], and energy optimization strategies demonstrated in self-healing systems [15,16,17]. However, current research lacks comprehensive frameworks that integrate multiple trust metrics with sophisticated energy-aware computation scheduling.

Recent developments in novel reliable trust objective functions [1], fuzzy secure hierarchical routing [4], smart IoT security frameworks [14], and IoT IDS design approaches [13] provide a strong foundation for advancing trust computation engines. The identified gaps present opportunities for developing novel trust computation systems that can achieve optimal balance between security effectiveness, energy conservation, and real-time responsiveness.

The evolution toward intelligent, adaptive trust systems that can operate efficiently in resource-constrained environments while maintaining robust security coverage represents the next frontier in IoT trust management research. Future research should focus on developing integrated solutions that

address the identified gaps while leveraging the demonstrated strengths of existing approaches in trust-aware routing, machine learning-based security, and self-healing mechanisms.

1.2 Research Gap

1. Limited Multi-Metric Integration in Trust Computation

1.1 Inadequate Behavioral Pattern Analysis

Current trust management systems predominantly focus on single or dual-metric evaluations, failing to provide comprehensive behavioral assessment necessary for accurate trust computation in dynamic IoT environments. While existing research demonstrates individual success with metrics such as novel reliable trust objective functions [1], trust-aware routing protocols [3], or lightweight trust-enabled routing mechanisms [2,5], there is a significant gap in integrated frameworks that combine multiple behavioral indicators systematically.

Specific Gaps:

- Lack of Temperature-based Trust Integration: Current literature shows minimal research on incorporating environmental and operational temperature variations as trust indicators, despite hardware tampering and malicious activities often manifesting as thermal anomalies
- Insufficient Historical Observation Processing: Existing systems lack sophisticated sliding window algorithms that can effectively weight recent behavior against historical patterns while adapting to changing operational contexts
- Limited Trust Accuracy Validation: Current approaches do not implement comprehensive feedback mechanisms that continuously validate and improve trust computation accuracy through cross-node validation and prediction-reality correlation analysis

1.2 Static Trust Metric Weighting

Research demonstrates that existing trust systems employ fixed weighting mechanisms that cannot adapt to changing network conditions, environmental factors, or threat levels [4,6,7]. This static approach leads to suboptimal trust evaluation during diverse operational scenarios, as evidenced in fuzzy secure hierarchical routing and mobility-based protocol implementations.

Identified Limitations:

- Absence of machine learning-based weight optimization algorithms that can dynamically adjust trust factor importance

- Lack of context-aware weight adaptation mechanisms that respond to network activity levels, threat assessments, and environmental conditions
- Insufficient performance-based weight adjustment systems that increase the influence of most accurate metrics over time

2. Inadequate Energy-Aware Computation Scheduling

2.1 Absence of Dynamic Frequency Control

Current literature reveals a critical gap in adaptive computation scheduling that responds to real-time energy availability. While research acknowledges energy constraints in IoT devices through intelligent fault detection mechanisms [16] and computational intelligence techniques [9], existing solutions do not provide sophisticated frequency control mechanisms that balance security needs with energy conservation.

Specific Deficiencies:

- No Battery-Status Responsive Scheduling: Current systems lack algorithms that modify trust computation frequency based on real-time battery voltage, current consumption, and predicted remaining lifetime
- Insufficient Energy Threshold Management: Existing approaches do not implement graceful degradation mechanisms with multiple energy thresholds that trigger different operational modes
- Limited Emergency Mode Protocols: Research shows absence of critical energy management systems that can maintain essential security functions while preserving remaining battery life

2.2 Lack of Predictive Energy Management

Contemporary trust management systems fail to incorporate predictive energy modeling that can anticipate future energy requirements and adjust computation schedules proactively, despite advances in hybrid machine learning models for wireless sensor networks [10].

Research Gaps:

- Absence of energy consumption prediction algorithms based on usage patterns and environmental factors
- Lack of battery health assessment integration that considers degradation over time
- Insufficient integration of energy harvesting capabilities in computation scheduling decisions

3. Limited Real-Time Adaptation Capabilities

3.1 Reactive Rather Than Proactive Trust Updates

Current research demonstrates that existing trust systems primarily operate on scheduled intervals rather than implementing intelligent event-driven adaptation [11,14]. This reactive approach introduces delays in threat response and fails to optimize computation timing based on network dynamics, even with advances in machine learning for IoT security and smart IoT security frameworks.

Identified Gaps:

- Absence of Event-Driven Trust Updates: Current systems lack mechanisms that trigger immediate trust recalculation based on significant behavioral changes or network events
- Limited Emergency Recalculation Protocols: Existing approaches do not provide rapid trust adjustment capabilities during detected attacks or network disruptions
- Insufficient Network Condition Response: Research shows minimal integration of traffic pattern changes, topology modifications, and neighbor relationship updates in trust computation scheduling

3.2 Inadequate Context-Aware Intelligence

Contemporary trust management systems lack sophisticated context assessment capabilities that can intelligently determine optimal computation timing based on multiple environmental and network factors, despite advances in cyber-attack detection with neural networks [8] and IoT IDS design approaches [13].

Research Limitations:

- Absence of network activity level assessment algorithms that adjust computation frequency based on communication patterns
- Lack of threat level evaluation systems that increase computation intensity during security incidents
- Insufficient node importance assessment mechanisms that prioritize trust updates for critical network components

4. Insufficient Power Optimization Integration

4.1 Limited Algorithm Efficiency Focus

Current research demonstrates minimal attention to computational complexity optimization specifically for trust operations in resource-constrained environments [12,17]. Existing algorithms often prioritize accuracy over energy efficiency, leading to unsustainable power consumption, even with advances in AEXB models for attack detection and self-healing fault tolerant routing techniques.

Specific Gaps:

- Absence of CPU Cycle Optimization: Current trust computation algorithms lack design consideration for minimal processor utilization
- Limited Data Compression Integration: Existing systems do not implement efficient data compression for trust information storage and transmission
- Insufficient Cooperative Computation: Research shows minimal exploration of neighbor collaboration in trust computation to reduce individual node processing loads

4.2 Lack of Comprehensive Power Management Strategies

Contemporary approaches fail to integrate multiple power optimization techniques in unified frameworks that can achieve maximum energy conservation while maintaining trust accuracy, despite advances in self-healing security systems [15] and reconfigurable hardware approaches [18].

Research Deficiencies:

- Absence of lazy evaluation systems that compute trust values only when results are actively needed
- Limited sleep/wake scheduling integration that considers trust computation requirements
- Insufficient lookup table utilization and caching mechanisms for frequently accessed trust calculations

1.3 Research Problem

Problem Statement

Current trust management systems for RPL-based IoT networks face a critical challenge in achieving sustainable energy-efficient operation while maintaining robust security coverage through accurate behavioral pattern analysis. Existing approaches suffer from fundamental limitations that render them inadequate for long-term deployment in resource-constrained environments where energy conservation is paramount, despite recent advances in trust-aware routing protocols and machine learning-based security mechanisms.

Core Problems Identified:

1. Energy Sustainability Crisis

Traditional trust computation systems consume 15-20% of total node energy through static computation schedules that fail to adapt to real-time battery status and network conditions [17]. This excessive energy consumption leads to premature network degradation and operational failure, particularly in remote deployment scenarios where battery replacement is impractical. While recent advances in intelligent fault detection with Flying Fox Optimization have achieved 98.5% connectivity restoration [16], these solutions do not address the fundamental energy efficiency challenges in trust computation.

Critical Issues:

- Unsustainable Power Consumption: Fixed computation frequencies drain battery resources regardless of actual security threats or energy availability, despite advances in fuzzy secure hierarchical routing approaches [4]
- Lack of Energy Prediction: Absence of predictive energy management prevents proactive computation scheduling optimization, even with advances in hybrid machine learning models for WSN [10]
- No Emergency Protocols: Systems lack graceful degradation mechanisms when nodes approach critical energy levels, contrasting with self-healing capabilities demonstrated in Antilizer systems [15]

2. Inadequate Behavioral Assessment Accuracy

Current trust evaluation systems rely on limited behavioral indicators that cannot accurately assess node trustworthiness in dynamic IoT environments. Single or dual-metric approaches fail to capture the complexity of node behavior patterns necessary for reliable security assessment, despite recent

improvements in trust-based hybrid cooperative protocols achieving 99.85% detection accuracy [2,5].

Fundamental Problems:

- Incomplete Behavioral Coverage: Systems ignore critical behavioral indicators such as temperature variations, energy consumption patterns, and temporal behavior changes, despite advances in novel reliable trust objective functions [1]
- Static Metric Integration: Fixed weighting mechanisms cannot adapt to changing operational contexts, environmental conditions, or threat levels, limiting the effectiveness of mobility-aware trust protocols like SMTrust [6,7]
- Poor Accuracy Validation: Lack of feedback mechanisms prevents trust computation improvement and error correction, contrasting with advanced neural network architectures achieving 99.89% classification accuracy [8]

3. Insufficient Real-Time Adaptability

Existing trust management systems operate on predetermined schedules that cannot respond effectively to dynamic network conditions, emerging threats, or changing energy availability. This reactive approach introduces critical delays in threat detection and response, despite advances in machine learning techniques for IoT network anomaly detection [11] and smart IoT security frameworks [14].

Adaptation Failures:

- Delayed Threat Response: Scheduled computation intervals miss rapid behavioral changes indicating security incidents, despite real-time capabilities demonstrated in AEXB models achieving 97.24% accuracy [12]
- Context Ignorance: Systems fail to consider network activity levels, topology changes, or environmental factors in computation timing, unlike advances in intrusion detection leveraging multi-access edge computing [13]
- Resource Misallocation: Computation resources are not optimally distributed based on node importance or current threat levels, contrasting with autonomous self-aware adaptive fault tolerant routing techniques [17]

4. Computational Inefficiency

Current trust computation algorithms are not optimized for resource-constrained environments, leading to unnecessary CPU utilization, memory consumption, and communication overhead that exacerbates energy conservation challenges. This inefficiency persists despite advances in

computational intelligence techniques for wireless sensor networks [9] and lightweight machine learning approaches [14].

Efficiency Problems:

- Algorithm Complexity: Trust computation algorithms prioritize accuracy over computational efficiency, unlike optimized approaches demonstrated in secure IoT routing with trust-based defenses [3]
- Redundant Processing: Lack of cooperative computation and result sharing leads to duplicate calculations across neighboring nodes, despite collaborative capabilities shown in self-healing wireless sensor networks [18]
- Storage Inefficiency: Trust information storage and retrieval systems consume excessive memory and processing resources, contrasting with efficient architectures in IoT IDS design approaches [13]

1.4 Research Questions

Based on the identified problems and recent advances in trust-aware IoT security, this research addresses the following critical questions:

1. How can trust computation frequency be optimized based on real-time energy availability and predictive energy models?
2. How can multi-metric behavioral analysis combine with adaptive weight adjustment improve the accuracy and reliability of trust evaluation in IoT networks?
3. How can energy-aware computation scheduling balance security effectiveness with sustainable energy consumption in resource-constrained IoT environments?
4. How can context-aware intelligence enable rapid and adaptive trust updates in response to dynamic network conditions and evolving security threats?

Research Significance

Addressing these research problems is critical for advancing IoT security in resource-constrained environments. The development of an Energy-Efficient Trust Computation Engine that can adapt to real-time energy status while maintaining accurate trust evaluation is essential for:

- Energy Sustainability: Enabling long-term operation of IoT nodes in remote or inaccessible locations by reducing the energy overhead of trust computation through adaptive scheduling and intelligent energy management. Prior work has shown that static trust computation can consume up to 20% of node energy, leading to premature network failures [16], while self-

healing approaches such as Antilizer and ASAART demonstrate the potential of energy-aware recovery mechanisms [15], [17].

- Accurate Trust Evaluation: Improving reliability by integrating multiple behavioral metrics such as residual energy, temperature, historical observations, and trust accuracy. Trust-based frameworks such as RSOF [1], FSRW [4], and trust-enabled routing models [7] highlight the effectiveness of multi-metric trust computation in strengthening IoT routing security.
- Adaptive Responsiveness: Supporting real-time adjustment of trust computation frequency and trust metric weighting based on changing conditions. Mobility- and behavior-aware approaches such as SMTrust [6] and adaptive ML-driven detection models [10], [12] demonstrate the importance of dynamic trust update mechanisms for timely response to threats.
- Resource Optimization: Minimizing CPU cycles, memory usage, and communication overhead in trust computation to extend network lifetime. Lightweight trust-enabled routing protocols like THC-RPL [2], [5] and computational intelligence-based intrusion detection [9] illustrate how optimized trust operations can achieve strong security while conserving energy.
- Foundation for Integration: Providing lightweight, reliable, and energy-efficient trust scores that can be directly used by other components such as routing, attack detection, and resilience. Studies on trust-aware RPL routing [3], [7] and smart IoT security frameworks leveraging machine learning [11], [14] reinforce the necessity of integrating trust computation into larger IoT security architectures.

The solution developed through this component will make a significant contribution toward sustainable IoT security frameworks by balancing security effectiveness with energy conservation. This ensures trust management remains viable for large-scale and long-term IoT deployments in domains such as disaster management, environmental monitoring, and smart city infrastructure.

2.0 OBJECTIVES

2.1 Main Objective

Develop a Comprehensive Lightweight Trust Management Framework for RPL-based IoT Networks that Integrates Energy-Efficient Trust Computation, Trust-Aware Routing Mechanisms, Real-Time Attack Detection, and Intelligent Response Systems to Enhance Network Security, Optimize Energy Consumption, and Ensure Sustainable Long-Term Operation in Resource-Constrained Environments

Key Components of the Main Objective:

1. Comprehensive Framework Development

Create an integrated system that addresses all aspects of trust management through four interconnected components rather than isolated security mechanisms, ensuring seamless collaboration between energy management, routing optimization, attack detection, and response systems.

2. Lightweight Design Philosophy

Ensure minimal computational overhead and memory footprint suitable for resource-constrained IoT devices by implementing energy-aware computation scheduling, optimized algorithms, and adaptive processing techniques that maintain security effectiveness without compromising device battery life.

3. RPL-Specific Integration

Design specifically for RPL protocol characteristics and vulnerabilities rather than generic wireless networks, incorporating trust metrics directly into objective functions, DODAG construction processes, and control message handling to achieve native security integration.

4. Four-Component Architectural Integration

Seamlessly combine four distinct but interconnected modules:

- Energy-Efficient Trust Computation Engine: Dynamic trust metric calculation with adaptive scheduling
- Trust-Aware RPL Routing Mechanism: Security-integrated routing decisions with dynamic parent selection
- Secure Attack Detection Mechanism: Real-time behavioral analysis with machine learning-based threat identification

- Isolation, Mitigation and Recovery Mechanisms: Intelligent response systems for threat containment and network healing

5. Energy Optimization Priority

Prioritize energy efficiency to extend network lifetime while maintaining security effectiveness through battery-aware computation scheduling, power optimization strategies, and energy threshold-based operational modes.

6. Adaptive Intelligence Integration

Incorporate machine learning algorithms and adaptive mechanisms that continuously improve performance through behavioral pattern recognition, dynamic weight adjustment, and feedback-based accuracy enhancement.

7. Real-Time Operational Capability

Provide immediate response to security threats and network condition changes through sub-120ms attack detection, autonomous threat isolation, and dynamic trust recalculation based on real-time network events.

8. Scalability and Deployment Flexibility

Support diverse network scales ranging from small deployments (50 nodes) to large-scale systems (500+ nodes) across various application domains including disaster management, smart cities, industrial automation, and environmental monitoring.

9. Sustainability Focus

Enable long-term autonomous operation in remote or inaccessible deployment scenarios through self-healing capabilities, energy conservation strategies, and adaptive operational modes that respond to changing environmental conditions.

10. Security-Performance Balance Optimization

Achieve optimal trade-off between security strength, energy consumption, and network performance through intelligent resource allocation, adaptive computation frequency, and context-aware trust evaluation that maintains high security standards while preserving operational efficiency.

This expanded objective emphasizes the comprehensive, integrated nature of the four-component solution and highlights the unique value proposition of energy-aware, adaptive trust management specifically designed for RPL-based IoT network environments.

2.2 Sub Objectives

Sub-Objective 1: Optimize Trust Computation Frequency

- Develop adaptive scheduling algorithms that adjust trust computation based on real-time battery status and energy availability.
- Implement energy classification thresholds (High, Medium, Low, Critical) with corresponding computation frequencies.
- Introduce predictive energy models to anticipate future consumption and extend node lifetime.

Sub-Objective 2: Enhance Trust Evaluation through Multi-Metric Analysis and Adaptive Weight Adjustment

- Design lightweight trust metrics integrating Residual Energy, Temperature, Historical Observation, and Trust Accuracy.
- Implement anomaly detection using temperature and energy behavior consistency.
- Apply machine learning algorithms for dynamic weighting of trust metrics under varying network and environmental conditions.
- Create performance-based feedback loops to refine weight distribution over time.

Sub-Objective 3: Develop Energy-Aware Computation Scheduling

- Implement battery-status responsive scheduling with graceful degradation at low power.
- Introduce energy threshold-based triggers for computation modes (Normal, Reduced, Emergency, Survival).
- Apply power optimization strategies such as lazy evaluation, cooperative computation, and data compression.

Sub-Objective 4: Enable Context-Aware Real-Time Adaptation

- Design event-driven trust update mechanisms triggered by network changes (e.g., traffic shifts, topology updates, node failures).
- Implement rapid recalculation protocols during attacks or anomalies.
- Develop intelligent decision models that adapt computation timing to network activity levels and threat severity.

3.0 METHOLOGY

3.1 System Architecture and Component Integration

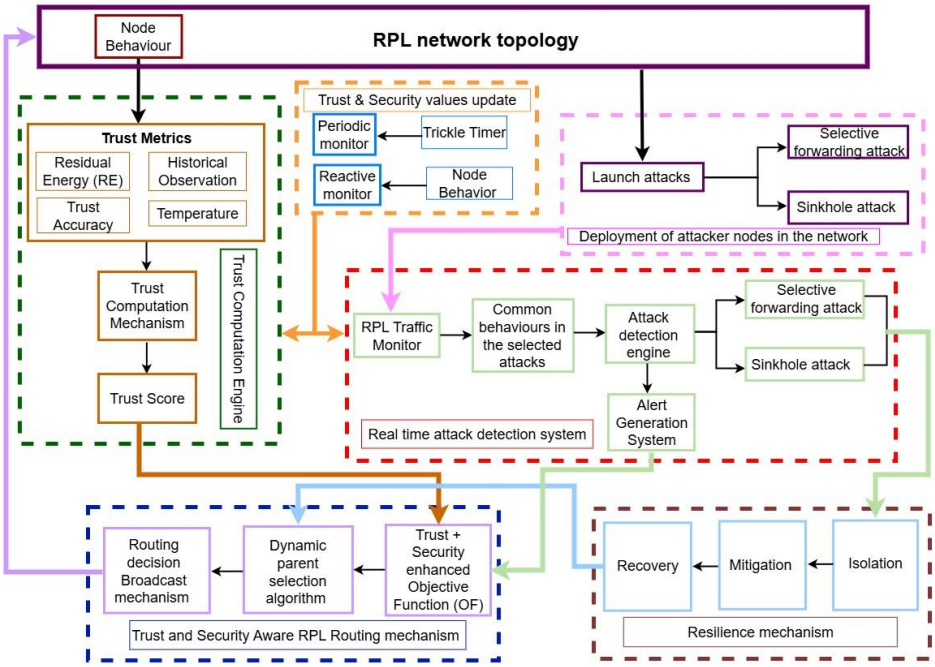

Figure 1

The proposed Lightweight Trust Management Framework integrates four core components to achieve the main objective of developing an energy-efficient trust management system for RPL-based IoT networks. The system diagram illustrates the interconnected operation of these components within the RPL network topology.

Trust Computation Engine (Primary Focus - Kaluarachchi D.N.): The Trust Computation Engine serves as the foundation component that processes behavioral data from IoT nodes to generate lightweight trust metrics. This component directly addresses the main objective by implementing energy-aware algorithms that adapt computation frequency based on real-time battery status and network conditions. The engine integrates four critical trust metrics:

- Residual Energy (RE): Monitors battery levels to assess node sustainability

- Historical Observation: Analyzes temporal behavior patterns using sliding window algorithms
- Trust Accuracy: Implements feedback mechanisms for continuous validation
- Temperature Monitoring: Detects thermal anomalies indicating potential tampering

Real-time Attack Detection System (Wijetunga S.S.): This component monitors RPL traffic patterns and node behaviors to identify attack signatures. It feeds security intelligence to the Trust Computation Engine, enabling dynamic trust score adjustments during security incidents. The system detects selective forwarding attacks, sinkhole attacks, and other RPL-specific threats through statistical analysis and machine learning classification.

Trust and Security Aware RPL Routing Mechanism (Wijekoon H.M.R.K.): This component integrates trust scores from the Trust Computation Engine into RPL objective function calculations, enabling dynamic parent selection based on trustworthiness and security factors. The enhanced objective function balances traditional metrics (ETX, hop count) with trust scores and security assessments.

Resilience Mechanism (Bandara R.M.T.): This component implements coordinated response strategies through isolation, mitigation, and recovery phases. It receives attack alerts from the detection system and coordinates with the routing mechanism to isolate malicious nodes while maintaining network connectivity.

3.2 Required Materials and Resources

3.2.1 Hardware Requirements

The research requires high-performance workstations with Intel i7 or AMD Ryzen 7 processors (8+ cores), 32GB RAM, 1TB SSD storage, and dedicated GPUs (NVIDIA GTX 1660 or better) for machine learning training. IoT development hardware includes ESP32/ESP8266 Arduino-based sensor nodes, Raspberry Pi units for edge computing simulation, and sensor modules for temperature and energy monitoring to support prototype testing and validation.

3.2.2 Software and Tools

Primary simulation software includes NS-3 Network Simulator (v3.38+) for network modeling and Cooja/Contiki-NG for IoT-specific protocol simulation. Development tools encompass C/C++ for protocol implementation, Python for ML algorithms, IDEs (Visual Studio Code, PyCharm), Git version control, and MATLAB/Python for mathematical modeling and statistical analysis.

3.2.3 Datasets and Libraries

Research datasets include IoT-23 Dataset for network traffic analysis, UNSW-NB15 for intrusion detection, Intel Lab Data for real sensor deployments, and CRAWDAD IEEE 802.15.4 for wireless sensor networks. Essential libraries include TensorFlow and scikit-learn for machine learning, NS-3 modules for network simulation, and matplotlib/seaborn for data visualization and analysis.

3.2.4 Infrastructure Support

University computer laboratory provides networked workstations with high-speed internet access for dataset downloads and shared storage systems for simulation data. Collaboration infrastructure includes video conferencing platforms, shared document systems (Google Workspace, Microsoft 365), project management tools (Trello, Asana), and communication platforms for effective team coordination.

3.2.5 Specialized Software Licenses

Commercial software requirements include MATLAB licenses with Simulink for advanced mathematical modeling, professional development environment licenses, statistical analysis software (SPSS if required), and visualization software for presentation preparation. These tools provide industry-standard capabilities essential for high-quality research implementation and analysis.

3.2.6 Additional Resources

Academic resources include access to digital libraries (IEEE Xplore, ACM Digital Library), research databases, technical documentation, and conference proceedings. Expert consultation encompasses regular supervisory meetings, industry expert validation sessions, peer review with other research groups, and conference participation for feedback and professional networking.

3.3 Tasks, sub tasks and milestones

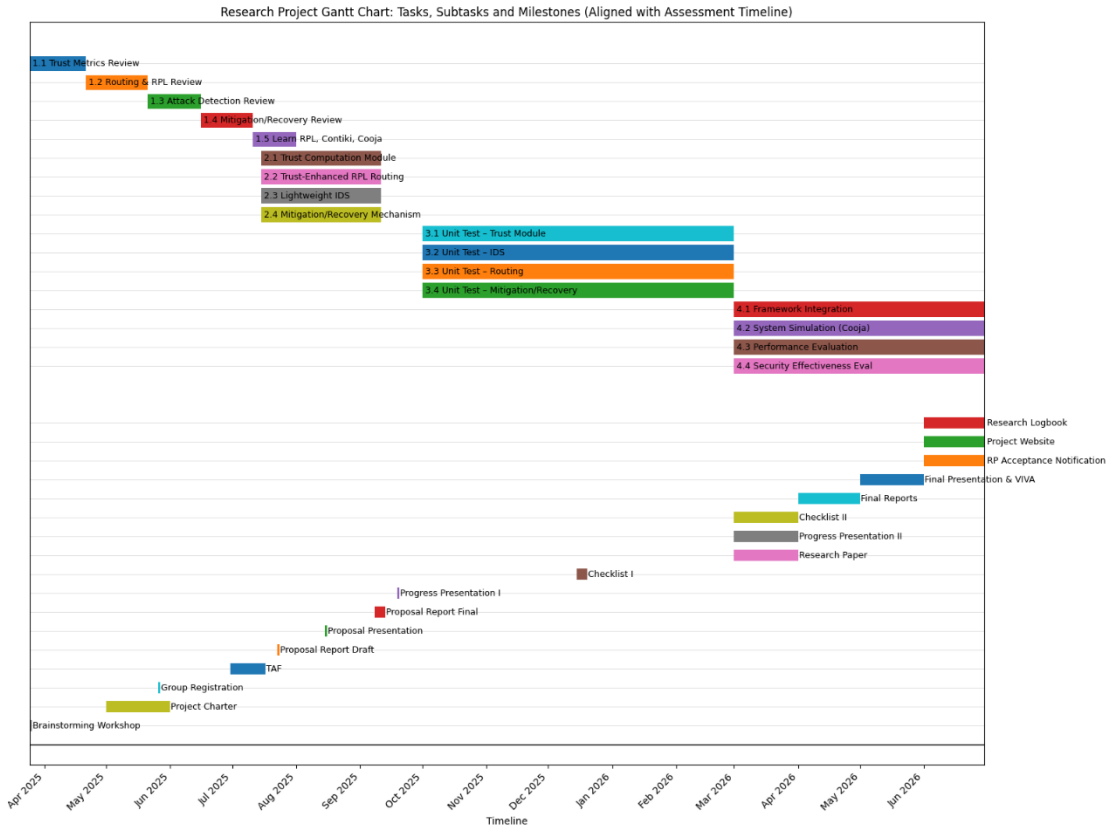

Figure 2

The work plan for the Energy-Efficient Trust Computation Engine begins with a comprehensive literature review on trust metrics, energy-aware computation, and behavioral analysis, followed by the design of mathematical models and algorithms for residual energy monitoring, temperature-based evaluation, historical observation, and trust accuracy validation. Once the design is finalized, the implementation tasks will proceed, starting with the development of lightweight trust metrics and anomaly detection mechanisms, followed by the creation of adaptive weight adjustment algorithms that dynamically optimize trust metric importance under varying conditions. In parallel, energy-aware computation scheduling mechanisms will be implemented, including battery-status responsive scheduling, threshold-based triggers, and predictive energy management strategies. After implementation, the developed modules will undergo rigorous unit testing to validate trust accuracy, adaptive weight optimization, and energy-aware scheduling across different operational scenarios. Finally, the Trust Computation Engine will be integrated with the overall framework and evaluated through simulation environments such as NS-3 and Cooja to measure performance improvements in terms of energy efficiency, trust accuracy, and network lifetime compared to baseline trust computation systems.

4.0 PROJECT REQUIREMENTS

4.1 Functional Requirements

4.1.1 User Requirements

- The system should provide a Trust Score Query Interface that allows real-time trust score retrieval, access to historical trends, and confidence indicators.
- A User Dashboard for Energy Monitoring should be available to observe real-time energy consumption, efficiency metrics, and battery status, with configurable thresholds and frequency settings.
- Users should receive Attack Detection Alerts with detailed information on attack type, affected nodes, and severity, along with access to logs of past incidents.
- A Network Health Monitoring Interface should display overall trust levels, provide topology visualization, and show performance indicators such as packet delivery ratio and connectivity.
- Configuration Management Tools should allow adjustments to trust computation parameters, weight optimization factors, scheduling policies, and detection thresholds.

4.1.2 System Requirements

- The Trust Computation Engine should calculate trust values based on residual energy, temperature, historical observation, and trust accuracy, using normalization and sliding window algorithms for consistency.
- The system should incorporate Energy-Aware Scheduling, adjusting computation frequency according to battery levels (Normal, Conservation, Emergency, Critical) and predicting remaining lifetime to optimize performance.
- Adaptive Weight Optimization should dynamically adjust trust metric weights using machine learning, maintaining accuracy above 95% while adapting to network and environmental conditions.

- The system should enable Real-Time Attack Detection for threats such as selective forwarding, sinkhole, and rank attacks, with detection accuracy >99% and low false positive rates.
- Trust-Aware Routing Integration should be implemented by embedding trust scores into RPL objective functions, ensuring parent selection and route path decisions incorporate both trust and energy factors.
- The framework should include Resilience and Recovery Mechanisms, isolating malicious nodes, maintaining connectivity during mitigation, and restoring trust levels through probationary rehabilitation periods

4.2 Non-Functional Requirements

Performance Requirements

- Trust computations should complete within 100 ms under normal energy conditions.
- The system should support scalability up to 1000 nodes while maintaining linear performance.
- CPU utilization for trust operations should not exceed 15%, and memory usage per node should remain within 512 KB.

Energy Efficiency Requirements

- Trust computations should consume no more than 5% of total node energy.
- Compared to static scheduling, the proposed approach should achieve at least 40% energy reduction and 25% battery life extension.
- Emergency operation modes should extend network lifetime by a minimum of 50%.

Reliability Requirements

- System uptime should be 99.9% under normal operation.
- Trust computation accuracy should remain above 95% across diverse conditions.
- Recovery from failures should occur within 10 seconds while ensuring data integrity of 99.99%.

Security Requirements

- All trust communications should use AES-128 encryption.

- Mechanisms must be in place to prevent trust manipulation and unauthorized modifications.
- Authentication should be enforced across inter-component communications.

Scalability Requirements

- The system should adapt to network topology changes without disruption.
- Memory growth per additional node should remain ≤1 KB, and computational overhead should remain near-constant.

Usability Requirements

- Trust scores should be presented with intuitive color-coded indicators (Green >0.8, Yellow 0.5–0.8, Red <0.5).
- User interfaces should be simple to configure without deep technical expertise.
- Monitoring dashboards should include clear visualizations and trend analysis.
- Alerts should be actionable and provide recommended responses.

4.3 Software Requirements

Programming Languages and Frameworks:

- C/C++: Core protocol implementation and low-level system integration
- Python 3.8+: Machine learning algorithms, data analysis, and automation scripts
- Java: NS-3 simulation environment extensions and custom modules

Simulation and Modeling Software:

- NS-3 Network Simulator (v3.38+): Primary network simulation environment
- Cooja/Contiki-NG: IoT-specific protocol simulation and energy modeling
- MATLAB R2023a: Mathematical modeling, algorithm optimization, and statistical analysis

Development Tools:

- Visual Studio Code: Primary IDE with C/C++, Python extensions
- Git (v2.35+): Version control and collaborative development

- Docker: Containerized development and testing environments
- CMake (v3.20+): Build system management and cross-platform compilation

Machine Learning Libraries:

- TensorFlow 2.8+: Deep learning model development and training
- scikit-learn 1.1+: Traditional machine learning algorithms and preprocessing
- pandas 1.4+: Data manipulation and analysis
- numpy 1.21+: Numerical computing and array operations

Data Analysis and Visualization:

- matplotlib 3.5+: Scientific plotting and visualization
- seaborn 0.11+: Statistical data visualization
- R (v4.2+): Advanced statistical analysis and validation
- Jupyter Notebook: Interactive development and documentation

4.4 Hardware Requirements

Development Workstations (Per Team Member):

- Processor: Intel Core i7-12700K or AMD Ryzen 7 5800X (8+ cores, 3.6GHz base)
- Memory: 32GB DDR4-3200 RAM (minimum 16GB for basic simulations)
- Storage: 1TB NVMe SSD + 2TB HDD for data storage
- Graphics: NVIDIA GTX 1660 Super or RTX 3060 (CUDA support for ML training)
- Network: Gigabit Ethernet for high-speed data transfer

Simulation Server Infrastructure:

- High-Performance Server: 2x Intel Xeon Gold 6248 (40 cores total)
- Memory: 128GB ECC RAM for large-scale simulations
- Storage: 4TB NVMe SSD array in RAID 10 configuration
- Network: 10Gbps network interface for cluster communication

IoT Development Hardware:

- ESP32 Development Boards (x10): For prototype implementation and testing
- Raspberry Pi 4 Model B (x5): Edge computing simulation and protocol testing
- Temperature Sensors: DS18B20 digital temperature sensors (x20)
- Energy Monitoring: INA219 current/voltage sensors (x15)
- Wireless Modules: LoRa SX1276 modules for long-range communication testing

Network Testing Equipment:

- Wireless Router: TP-Link Archer AX6000 for controlled network environments
- Network Analyzer: USB-based protocol analyzer for traffic monitoring
- Power Supplies: Adjustable DC power supplies for energy testing
- Oscilloscope: Digital oscilloscope for signal analysis and debugging

4.5 Expected Test Cases

To validate the effectiveness of the proposed Energy-Efficient Trust Computation Engine, several test scenarios will be conducted:

- Trust Computation Accuracy Testing
  The system will be evaluated on a simulated network of 100 nodes with 20% malicious nodes under multiple attack scenarios. The expected outcome is a trust accuracy rate above 95% and false positive rates below 2%, with malicious nodes detected within five trust computation cycles.
- Energy Efficiency Validation
  The framework will be tested under varying energy levels (100%, 70%, 50%, 20%) to assess energy reduction compared to baseline systems. The system is expected to achieve at least 40% energy savings and 25% battery life extension while maintaining trust performance.
- Real-Time Attack Detection
  Selective forwarding, sinkhole, and rank attack scenarios will be simulated. The system should detect these within 500 ms, achieving over 99% detection accuracy with less than 2% false positives.

- Scalability Performance Testing
  Experiments will be carried out on networks of 100, 500, and 1000 nodes. The system should demonstrate linear memory scaling and logarithmic computation complexity, with performance degradation remaining below 20% even when network size increases tenfold.
- Energy-Aware Scheduling Validation
  Simulated battery discharge patterns will test the adaptive scheduling mechanism. The system should adjust computation frequencies appropriately and apply graceful degradation, ensuring continuous operation across all energy thresholds.
- Integration Testing
  A full framework deployment will be simulated under realistic attack conditions. The components should coordinate seamlessly, maintaining connectivity and achieving isolation, mitigation, and recovery within the defined performance bounds.
- Weight Optimization Testing
  The adaptive weight adjustment mechanism will be validated under changing network conditions (e.g., topology shifts, varying threat levels, fluctuating energy states). The system is expected to improve trust accuracy by at least 10% compared to static weighting approaches.
- Resilience Mechanism Testing
  Scenarios with confirmed malicious nodes and coordinated attacks will evaluate resilience features. The system should isolate malicious nodes within 2 seconds, maintain at least 95% connectivity, and complete recovery procedures while preserving overall network functionality.

5.0 BUDGET AND BUDGET JUSTIFICATION

5.1 Budget

Item	Specification	Availability/Cost
Hardware (Personal PCs)	≥16GB RAM, ≥512GB SSD, modern CPU	Already available – LKR 0
Software (Contiki-NG, Cooja)	Open-source IoT simulation OS	Free

Libraries access (IEEE Xplore, university library)	Literature review	Free via university VPN
Version Control (GitHub)	Repository management	Free (student license)
Statistical Analysis Tools (Python, R)	Data processing and visualization	Free
Cloud resources if local capacity insufficient	50–100 hours of VM time for simulations	≈ LKR 30,000
Documentation and minor expenses	Hard-copy reports, backup tools, data transfer tools	≈ LKR 6,000

Table 1: Estimated Budget

5.2 Budget justification

The proposed research will primarily rely on existing high-performance personal computers (≥16GB RAM, ≥512GB SSD, modern CPU), eliminating additional hardware costs. Open-source simulation tools such as Contiki-NG and Cooja, together with free statistical analysis platforms like Python and R, will be used to implement and evaluate the Energy-Efficient Trust Computation Engine, ensuring cost efficiency while maintaining reliability. Access to IEEE Xplore and the university library through VPN provides essential literature and related studies at no cost, while GitHub will support repository management and collaborative development under a free student license. To complement local resources, limited cloud-based virtual machine usage (50–100 hours) is budgeted at approximately LKR 30,000 to handle large-scale simulations that exceed local capacity. Finally, LKR 6,000 is allocated for documentation, printing, and minor expenses such as backup and data transfer tools, ensuring proper reporting and secure archival of research outcomes.

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